Notes: 1) The distribution of $\bar{X}$ gets "normal" when $n \approx 30$

2) $\mu_{\bar{X}} = E(\bar{X}) = \mu = 3.5$

3) $\sigma_{\bar{X}}^2 = Var(\bar{X}) = \frac{\sigma^2}{n} = \frac{2.9167}{n}$

Recall: Suppose $X_1 \ldots X_n$ is a Random Sample from a <u>Normal</u> distribution with mean μ and variance σ^2

$X_i \overset{iid}{\sim} N(\mu, \sigma^2)$ $i = 1, \ldots n$ then

$\bar{X} \sim N(\mu_{\bar{X}} = \mu, \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n})$

<u>Theorem</u>. Central limit theorem (CLT)

Suppose $X_1 \ldots X_n$ is a R.S. from a <u>non-normal</u> distribution ~~when~~ with mean μ, and variance σ^2 ($\sigma^2 < \infty$). If <u>$n \geq 30$</u>, then

$\bar{X} \dot{\sim} N(\mu_{\bar{X}} = \mu, \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n})$.

$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \dot{\sim} Z$ [standardization]

Theorem: Law of Large numbers

As $n \to \infty$, $\bar{X} \to \mu$.

Example: Verbal SAT scores have a $N(\mu = 500, \sigma^2 = 100^2)$

Suppose 4 people are randomly selected.

Find probability mean SAT of these 4 people is less than 450.

X_i iid. $N(500, 100^2)$.

$\bar{X} \sim N\left(\mu_{\bar{X}} = 500,\ \sigma^2_{\bar{X}} = \frac{\sigma^2}{n} = \frac{100^2}{4} = 50^2\right)$

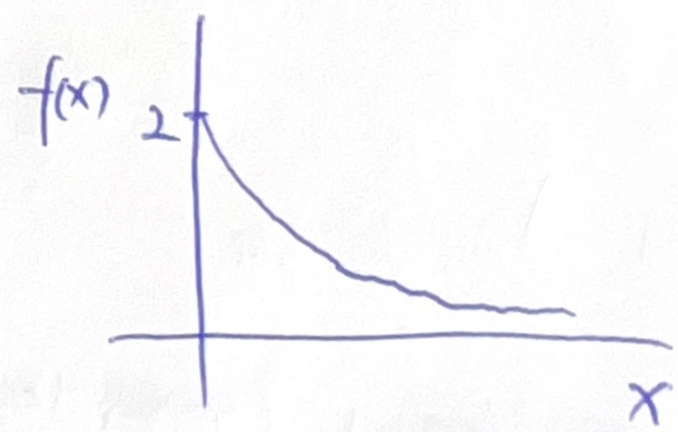
$P(\bar{X} < 450) = P\left(\boxed{\frac{\bar{X} - 500}{50}} < \frac{450 - 500}{50}\right)$

$= P(Z < -1) = 0.1587$.

<u>Example</u>: Machine error, X of a randomly selected part can be modeled by a exponential distribution with $\lambda = 2$.

$X \sim \exp(\lambda = 2)$

$$f(x) = \begin{cases} 2e^{-2x} & x > 0 \\ 0 & \text{otherwise.} \end{cases}$$



$E(X) = \mu = \frac{1}{\lambda} = \frac{1}{2}$

$Var(X) = \sigma^2 = \frac{1}{\lambda^2} = \frac{1}{4} = \left(\frac{1}{2}\right)^2$

a). Randomly select a part, Find probability mach error < 0.3

$P(X < 0.3) = \int_{-\infty}^{0.3} f(x)\,dx = \int_0^{0.3} 2e^{-2x}\,dx = \underline{0.451}$

b) R.S. <u>100</u> parts, Find probability mean mach error $\bar{X} > 0.65$

Based on CLT, $\bar{X} \dot{\sim} N\left(\frac{1}{2}, \frac{1}{400} = \left(\frac{1}{20}\right)^2\right)$

$P(\bar{X} > 0.65) = P\left(\boxed{\frac{\bar{X} - \frac{1}{2}}{\frac{1}{20}}} > \frac{0.65 - \frac{1}{2}}{\frac{1}{20}}\right)$

$= P(Z > 3)$

$= 1 - P(Z \leq 3)$

$= 1 - 0.9987$

$= 0.0013$

$$P(\bar{X} > 0.65) = P\left(\frac{\bar{X} - \frac{1}{2}}{\frac{1}{2}/\sqrt{100}} > \frac{0.65 - \frac{1}{2}}{\frac{1}{2}/\sqrt{100}}\right)$$

c). Suppose 4 parts are R.S. Find $P(\bar{X} < 0.4)$.

$\bar{X} \sim ??$ Cannot use CLT.

<u>Theorem</u>: CLT for sums.

Suppose $X_1 \ldots X_n$ is a RS from a <u>non-normal</u> distribution with mean μ, and variance σ^2 $(\sigma^2 < \infty)$. <u>If $n \geq 30$</u>.

$$Y = X_1 + \cdots + X_n \overset{\cdot}{\sim} N(\mu_Y = n\mu,\ \sigma_Y^2 = n\sigma^2)$$